

CEAHLAU TOACA, Romania

WMO: 151080

Lat: 46.9775N

Lon: 25.9500E

Elev: 1899

StdP: 80.50

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-17.4	-25.3	0.5	-6.7	-21.5	0.7	-11.4	34.4	-6.9	34.2	-6.9	9.1	50	0.948

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	20.0	15.4	18.3	14.5	16.8	13.6	16.7	18.6	15.4	17.1	14.2	15.9	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	14.4	17.6	14.7	13.2	16.3	13.6	12.3	15.2	54.8	18.7	50.5	17.1	46.8	16.0	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.8	27.9	20.3	DB	-22.1	22.6	2.3	1.9	-23.8	23.9	-25.1	25.0	-26.4	26.0	-28.1	27.4	(4)
(5)			WB	-22.2	18.3	2.3	1.8	-23.8	19.6	-25.2	20.7	-26.5	21.7	-28.1	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-7.8	-7.3	-5.3	0.4	5.9	9.4	11.2	11.4	6.7	2.5	-1.5	-6.2	(6)
(7)		DBStd	8.39	5.23	5.04	4.91	4.61	4.31	3.74	3.55	3.72	4.41	5.32	5.70	5.28	(7)
(8)		HDD10.0	3253	552	485	476	288	139	56	29	28	117	237	344	502	(8)
(9)		HDD18.3	6089	810	718	734	538	387	268	222	217	351	491	594	760	(9)
(10)		CDD10.0	206	0	0	0	1	11	37	66	71	17	4	0	0	(10)
(11)		CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)
(12)		CDH23.3	1	0	0	0	0	0	0	0	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.7	9.8	9.9	8.8	6.9	6.1	5.5	6.3	5.4	6.6	7.1	9.5	10.1	(14)
(15)	Precipitation	PrecAvg	651	20	23	33	50	76	109	102	85	52	43	27	30	(15)
(16)		PrecMax	1027	40	44	65	100	160	282	200	156	129	110	57	70	(16)
(17)		PrecMin	470	7	8	12	15	32	39	43	28	17	4	0	3	(17)
(18)		PrecStd	114	8	8	13	22	35	53	40	35	25	23	14	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	6.2	7.2	13.6	17.5	20.5	21.8	22.3	19.2	15.9	11.2	6.5	(19)
(20)			MCWB	0.6	1.4	2.8	8.7	12.7	15.7	16.7	16.4	12.6	10.1	5.7	1.0	(20)
(21)		2%	DB	1.9	2.4	4.8	10.4	15.0	18.2	19.7	20.0	16.2	13.1	9.5	3.9	(21)
(22)			MCWB	-0.1	0.5	2.3	6.7	11.5	14.5	15.9	15.2	12.0	8.4	4.4	0.5	(22)
(23)		5%	DB	0.4	0.6	2.8	8.2	13.2	16.2	18.0	18.2	14.2	11.2	7.7	2.2	(23)
(24)			MCWB	-0.9	-0.7	1.0	5.3	10.4	13.3	14.9	14.5	11.2	7.3	3.7	0.0	(24)
(25)		10%	DB	-0.9	-0.7	1.1	6.2	11.7	14.7	16.3	16.5	12.5	9.4	6.2	0.6	(25)
(26)			MCWB	-1.9	-1.6	0.0	4.3	9.4	12.5	13.9	13.5	10.2	6.8	2.9	-0.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	3.1	4.6	10.2	14.4	17.6	18.4	18.4	14.5	12.6	8.1	3.5	(27)
(28)			MCDWB	2.8	3.8	5.7	12.4	15.7	19.1	20.0	20.4	16.6	14.7	9.5	4.3	(28)
(29)		2%	WB	0.6	0.9	2.8	7.3	12.4	15.5	16.7	16.5	12.9	10.0	6.3	2.1	(29)
(30)			MCDWB	1.2	2.0	4.1	9.3	14.0	16.9	18.6	18.6	15.2	11.4	7.8	2.8	(30)
(31)		5%	WB	-0.5	-0.2	1.3	5.9	11.0	14.0	15.5	15.1	11.6	8.5	5.1	0.7	(31)
(32)			MCDWB	0.1	0.5	2.4	7.6	12.5	15.6	17.3	17.0	13.5	9.9	6.6	1.3	(32)
(33)		10%	WB	-1.8	-1.4	0.1	4.6	9.8	12.8	14.3	13.9	10.4	7.2	3.8	-0.4	(33)
(34)			MCDWB	-1.3	-0.9	1.0	6.0	11.2	14.3	15.8	15.8	12.1	8.7	5.2	0.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	5.0	5.4	5.6	6.4	6.5	6.5	6.4	5.8	5.3	4.9	4.6	(35)
(36)			MCDBR	5.0	5.8	6.8	7.7	8.1	8.4	8.4	7.8	7.6	6.6	5.7	5.1	(36)
(37)			MCWBR	4.2	5.1	5.3	5.6	5.9	6.2	6.2	5.8	5.8	5.1	4.3	3.9	(37)
(38)		5% WB	MCDBR	4.6	5.5	6.1	7.0	7.7	8.0	7.6	7.1	6.2	5.4	4.5	4.5	(38)
(39)			MCWBR	4.4	5.1	5.1	5.6	6.3	6.5	6.4	6.2	6.0	5.4	4.8	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.243	0.265	0.296	0.339	0.325	0.324	0.340	0.345	0.314	0.287	0.267	0.245	(40)	
(41)		taud	2.455	2.425	2.430	2.360	2.441	2.456	2.403	2.389	2.466	2.542	2.550	2.495	(41)	
(42)		Ebn at Noon	878	921	935	916	939	938	917	899	903	884	843	836	(42)	
(43)		Edh at Noon	64	82	96	114	109	108	113	109	92	73	59	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.24	2.04	3.19	4.33	5.42	5.76	5.75	5.26	3.73	2.48	1.35	0.96	(44)	
(45)		RadStd	0.17	0.25	0.36	0.41	0.49	0.49	0.49	0.44	0.43	0.31	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.81	N/A	N/A	N/A	N/A	N/A	-243	-303	+51	N/A	(46)
(47) Regional (55 neighbors)		+0.69	N/A	N/A	+0.82	N/A	N/A	-158	-220	+83	+19	(47)

Nomenclature: See separate page